

Technology Stack

Date	06 July 2026
Team ID	
Project Name	Smart Lender - Loan Eligibility Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	<i>HTML, CSS, Bootstrap, javascript</i>	<i>Creates a responsive and user friendly interface for entering loan details and displaying prediction results.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Handles user requests, processes input data, and communicates with the machine learning model efficiently.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	SQLite, CSV Dataset, Joblib (.pkl)	Stores user information, dataset, and trained machine learning model for prediction.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render / PythonAnywhere	Provides easy deployment, online accessibility, and cost-effective hosting.
5	Version Control & CI/CD	Git, GitHub	Tracks code changes, supports team collaboration, and maintains project versions.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Pandas, NumPy, Scikit-learn, XGBoost</i>	<i>Used for data preprocessing, model training, feature engineering, and loan eligibility prediction.</i>